Vipul Sapra

Charlotte, North Carolina | 513-693-6897 | vpsapra5@gmail.com | linkedin.com/in/vipul-sapra-ab30a09a

EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Generative AI & LLMs, Agentic AI, Autonomous Workflows, Retrieval-Augmented Generation (RAG), MLOps, LLMOps, Predictive Machine Learning, Time-Series Forecasting (SARIMA), Autonomous Anomaly Detection, Distributed Data Processing, Dimensional Modeling (Star/Snowflake Schemas, SCD Type 1/2), Data Governance & Quality, Enterprise Semantic Modeling

Programming Languages & Tools: Snowflake, Snowflake Cortex AI, AWS (S3, Redshift), Azure (ADF, Databricks, Synapse, Azure SQL), Apache Spark, PySpark, Python, XGBoost, Scikit-Learn, KIRO IDE, GitHub Copilot, Power BI (DAX), Tableau, Collibra, SQL

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior Snowflake Forward Deployed AI & Data Engineer — **May 2022 – Present**

- **Architected and deployed custom autonomous agentic workflows** (utilizing KIRO IDE, prompt orchestration, and Copilot) to automate schema discovery, data profiling, regex classification, and code migration, reducing manual delivery turnaround by 40-50% for enterprise clients.
- **Engineered production-grade GenAI-enabled solutions** and automated testing/evaluation frameworks for enterprise virtual assistants like SpectrumIQ, cutting manual QA turnaround by 50% and accelerating production release cycles in fast-paced environments.
- **Developed autonomous AI anomaly detection frameworks** identifying schema variations, distribution drift, and ETL discrepancies with 95% precision using advanced machine learning, safeguarding downstream BI, data pipelines, and cloud platform integrity.
- **Built distributed cloud data platforms and dimensional models** (Star/Snowflake schemas, SCD Type 1/2) leveraging Snowflake, Azure (ADF, Databricks, Synapse), AWS (S3, Redshift), and PySpark to process hundreds of terabytes while maintaining 99.9% data availability SLAs.
- **Collaborated directly with senior stakeholders and engineering pods** to translate complex business requirements into high-impact AI prototypes, executive Key Operating Metric (KOM) dashboards, and data governance frameworks using Collibra and Power BI.

Data Scientist & Machine Learning Specialist — **May 2022 – Present**

- **Deployed scalable XGBoost, Random Forests, and Scikit-Learn models** predicting complex customer behavior and delivering actionable behavioral signals that improved operational service reliability by 20%.
- **Architected advanced time-series forecasting models (SARIMA)** to predict truck rolls and field support demand, optimizing workforce capacity allocation and reducing response times by 20%.
- **Engineered source-to-target reconciliation firewalls** and robust automated data quality frameworks that slashed operational turnaround times by over 80% across core enterprise systems.

Business Intelligence & Decision Analytics Specialist — **May 2022 – Present**

- **Designed enterprise-grade semantic models and executive dashboards** in Power BI (DAX) and Tableau, successfully translating complex transactional telemetry into actionable corporate growth strategies.
- **Optimized high-throughput SQL and data transformation pipelines** ensuring high performance and low latency for critical downstream analytics reporting and executive decision intelligence.
- **Partnered with cross-functional teams to define KPI metrics** and streamline data governance policies, aligning analytical outputs with core organizational business drivers.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture **Pune, India**

Test Engineering Analyst **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.